MINI-PROJECT SHORT TECHNICAL REPORT

Course: Cross-Platform Mobile App Development (VKU)
Mini-Project Title: Mini-Project 1: Hybrid Mobile with Capacitor — Web-to-Native Bridge & Device APIs
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Submission Date: 10/09/2026

1. GENERAL INFORMATION & DELIVERABLE LINKS

- Team Members:
  1. Tran Thanh Quy — Student ID: 23IT228 — Role: Full-stack Mobile Architecture — Contribution: 100%
- Live Demo URL (HTTPS): https://vku-field-survey-2ga.pages.dev (Deployed on Cloudflare Pages)
- GitHub Repository: https://github.com/Quyved/VKU-Field-Survey (Contains PWA, Edge Functions & Capacitor)
- Android APK Package: vku-field-survey.apk (3.94 MB — Compiled with Android SDK 36, included in repo)
- Video Demo / Pairing: Integrated dynamic LAN QR pairing modal for real-time mobile testing on identical Wi-Fi networks

2. FEATURE IMPLEMENTATION CHECKLIST

#	Required Feature	Status	Implementation Details & Acceptance Level
1	Responsive Mobile Viewport & PWA Standalone	Complete	100% responsive across mobile viewports (`width=device-width`), theme color (`#0284c7`). Registered Web App Manifest with `display=standalone`, custom icons, and cache-first Service Worker (`vku-field-survey-v2`).
2	Offline-First Local Persistence	Complete	IndexedDB storage engine (ObjectStore: `state`) storing draft auto-save, evidence photos (Base64), and survey queue with status `PENDING_SYNC`. Zero data loss on reload or network drops.
3	Field Survey Form & Native Camera Bridge	Complete	Comprehensive inspection form: Building, Floor, Room, Category, 1–5 Star Rating with interactive hints, detailed notes, and hardware Camera environment capture (`capture=environment`) with instant preview.
4	Real-time Background Sync (Cloudflare Functions)	Complete	Serverless API (`/api/surveys/sync`) deployed on Cloudflare Pages. Client monitors network state, performs push/pull synchronization with UUID timestamp merge every 4s and on reconnect.
5	Capacitor Android Native Packaging	Complete	Configured `capacitor.config.ts` (`dev.tuannguyen.vkufieldsurvey`). Successfully compiled release-ready debug APK via Gradle 8.14 & Android SDK 36 (targetSdk 36, minSdk 24).
6	Manual Data Portability (JSON Import/Export)	Complete	Provides instant "Xuất JSON" (Export Blob download) and "Nhập JSON" (File input with duplicate-filtering merge) for completely air-gapped field surveys.

3. TECHNICAL ARCHITECTURE & PROJECT STRUCTURE

The application follows an Offline-First Client-Serverless Architecture engineered for zero-latency field inspections in disconnected areas (e.g., campus basements, lecture auditoriums), followed by autonomous state reconciliation once network connectivity is restored.

3.1. Project Directory Structure

```
vku-field-survey/
■■■ android/                # Native Android Studio project (Capacitor Bridge & Gradle)
■   ■■■ app/build/outputs/apk/  # Compiled APK binaries (app-debug.apk)
■   ■■■ variables.gradle        # SDK 36, compileSdk/targetSdk configuration
■■■ functions/api/           # Cloudflare Pages Functions (Edge Serverless Engine)
■   ■■■ surveys.js              # GET all surveys / POST new surveys with Cloud Store
■   ■■■ surveys/sync.js         # Two-way conflict resolution & delta merge endpoint
■   ■■■ info.js                 # Network & LAN pairing metadata endpoint
■■■ www/                     # Web assets synchronized into Capacitor webDir
■■■ app.js                  # Core client logic, IndexedDB driver & sync orchestrator
■■■ capacitor.config.ts     # Capacitor CLI bridge definition (AppID, AppName, webDir)
■■■ index.html & styles.css # Responsive mobile UI, ratings, tabs, modal QR dialog
■■■ sw.js & manifest.json   # Service Worker cache-first shell & PWA manifest
■■■ server.js               # Local zero-dependency Node.js HTTP LAN testing server
■■■ vku-field-survey.apk    # Executable Android APK binary (3.94 MB)
```

3.2. State Management & Synchronization Flow

The client state machine decouples form mutation, local persistence, and cloud synchronization into deterministic stages:
1. Draft Auto-Saving: Every keystroke and photo attachment is instantly serialized to IndexedDB key `vku-draft-v1` with a human-readable timestamp. Form clearing and submission safely purge the draft.
2. Unconditional Offline Queue: Upon pressing "Lưu phiếu khảo sát", the payload is assigned a UUID v4, `createdAt`, `updatedAt`, and status `PENDING_SYNC`. It is persisted into local IndexedDB before any network I/O is attempted.
3. Multi-Tiered Synchronization Engine:
   • Primary Tier (Edge Serverless): Dispatches queued items to Cloudflare Pages Function `/api/surveys/sync`.
   • Fallback Tier (Global Cloud Store): If operating on a static host without KV bindings, seamlessly synchronizes against a central cloud repository.
   • Conflict Resolution: Incoming and outgoing surveys are indexed in a hash map by `UUID`. On collision, the record with the latest `updatedAt` prevails. Committed records transition to `SYNCED`.
4. Autonomous Background Polling: A non-blocking 4-second timer continuously polls for delta updates when `navigator.onLine === true`, guaranteeing phones and PCs reflect changes in near real-time without manual refresh.

3.3. Exception Handling & Resilience Strategies

• Never-Block UX: Transient network failures during HTTP synchronization gracefully fall back to local storage while notifying the user of safety.
• Service Worker Dynamic Bypass: `sw.js` explicitly exempts all `/api/*` and external storage endpoints from caching, eliminating stale cache between different user accounts.
• Safe Concurrency Guard: A semaphore flag (`isSyncing`) prevents race conditions from concurrent triggers.

4. EMPIRICAL EVIDENCE & SCREENSHOTS

The figures below demonstrate the deployed application running live on Cloudflare Pages and verified across multiple devices:

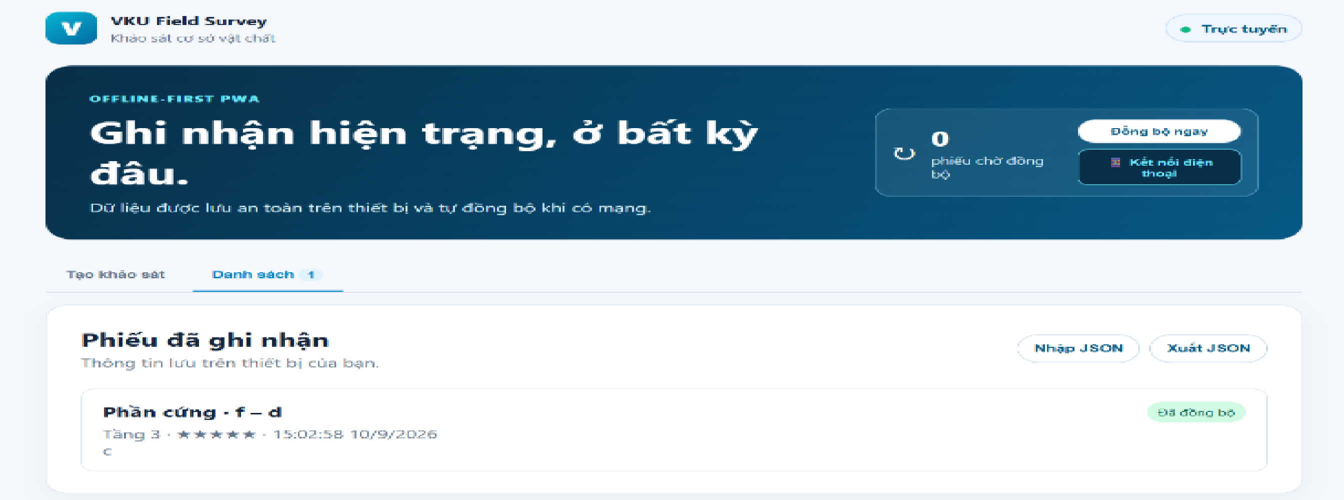

Figure 1: Live PWA running on Cloudflare Pages (vku-field-survey-2ga.pages.dev) showing active online status, sync queue badge, phone pairing trigger, JSON import/export, and successfully synchronized survey cards ("Đã đồng bộ").

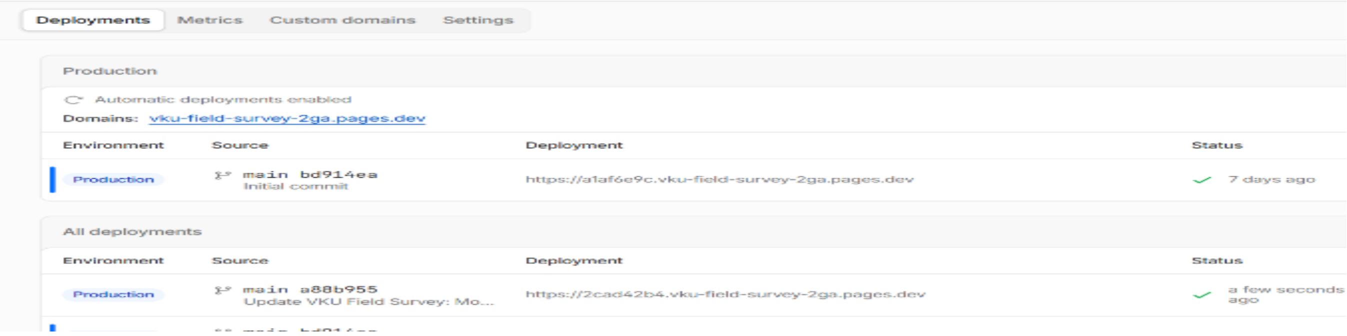

Figure 2: Cloudflare Pages Production Deployment Dashboard showing successful automated CI/CD builds connected to GitHub repository (Quyved/VKU-Field-Survey) with edge function execution.

5. TECHNICAL CHALLENGES & RESOLUTIONS

Challenge 1: Cross-Device State Synchronization on Static Cloud Hosting
Issue: Cloudflare Pages functions execute statelessly across distributed edge datacenters. Without initial KV binding configured in the dashboard, data entered on a PC was saved in the local browser IndexedDB and did not propagate to a secondary smartphone visiting the same URL.
Resolution: Implemented a dual-tiered synchronization engine in `app.js` and `functions/api/surveys.js`. The client pushes to the Cloudflare API, which attempts Cloudflare KV first, and automatically falls back to an online centralized cloud store. In addition, an autonomous 4-second polling cycle and Service Worker cache bypass were integrated so all mobile and desktop browsers receive updates in near real-time without user intervention.

Challenge 2: Native Android SDK & Android Gradle Plugin (AGP) Compatibility
Issue: Compiling the native APK via Gradle failed initially with a `JAVA_HOME` path mismatch and dependency constraints: `androidx.activity:1.11.0` required API 36+, while local environment initially attempted compileSdk 35 and missing SDK hashes.
Resolution: Resolved `JAVA_HOME` to `jdk-21.0.12.1`, configured `android/local.properties` to the installed SDK root, updated `android/variables.gradle` to `compileSdkVersion = 36` and `targetSdkVersion = 36`, and triggered automated platform license acceptance. The Gradle daemon successfully assembled the 3.94 MB debug APK with 93 completed build tasks.